ABSTRACT OF THE DISCLOSURE

A satellite communication system configured for high-radix physical layer encoding and parametric data transport over space-to-Earth and Earth-to-space links. The system comprises ground stations and satellite transponders equipped with high-radix RF transceivers capable of generating and distinguishing N discrete, distinguishable RF states (where N greater than 2) optimized for the specific challenges of satellite channels (e.g., free-space path loss, atmospheric effects, Doppler shift). Data is transmitted as unevaluated parametric expressions, ensuring end-to-end preservation of data precision across the satellite link, from source to destination, without loss of fidelity due to intermediate binary conversion or processing, and enabling the robust communication of perfect-precision symbolic data.